Design Details Document

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Design of Pass Robot (PR):-

Specifications:-

Length	1155mm (full ext.)
Width	1070mm (full ext.)
Height	872.5mm
Weight	Approximately 22kg

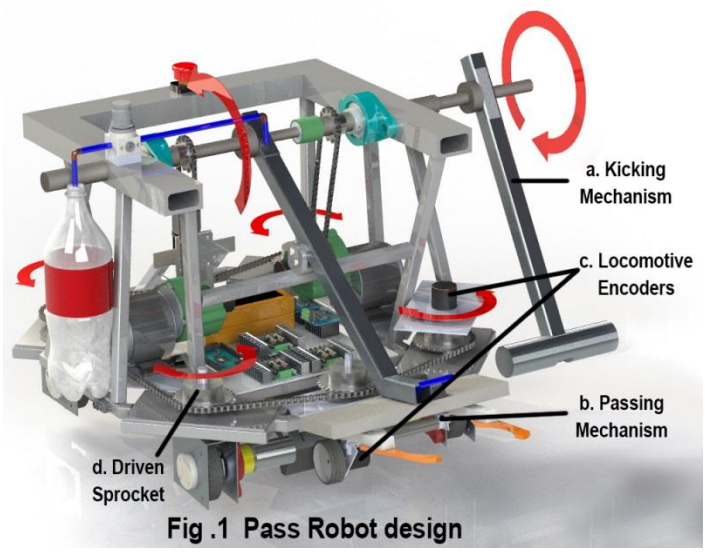
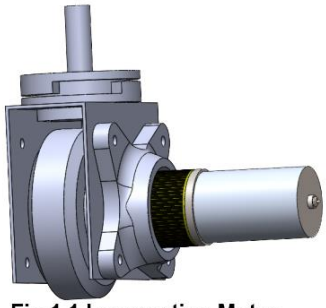


Fig .1 Pass Robot design

Type of drive:- (Synergic four wheel steering drive)

The PR has an Octagonal chassis (Fig 1.2) made of 20 x 20 x 2 mm 6061 grade Aluminium square tubes as Aluminium ensures sturdiness due to its lightweight and structurally rigidity.

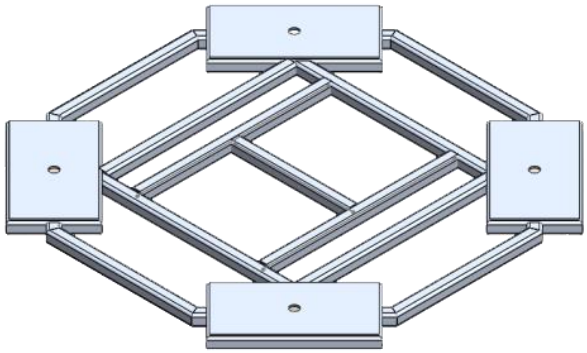
The Pass Robot (PR) uses a four wheel steerable drive which has superior maneuvering capabilities over other drives. Each of the four wheels has a planetary gear motor (which has low noise level, great durability and relatively high amount of torque transmission) that acts as the source of locomotive force. All the wheel casings are attached to sprockets which are connected through a common chain system which is sufficiently driven by a single Super Hercules motor. During translation, each wheel is powered individually to propel the robot while the direction of motion is controlled accurately by the Super Hercules motor and this motor needs to rotate only for half of its complete rotation to move in any direction and be present in any orientation.


Fig 1.1 Locomotion Motor

The locomotion of the PR is controlled through a PS2 controller which sends commands to the Arduino Due, the brain of the robot, and in return the Due issues commands to Hercules Motor drivers (Peak Current-13A) which drive the locomotion motors to achieve the desired motion. In contrast to that, the kicking and passing tasks are done autonomously making the PR a semi-autonomous robot.

From the comparison below, **the steering drive has an advantage** in terms of **speed** when compared to the other drives. Thus the synergic steering drive mechanism was selected due to its precise working, ease of operation, speed, acceleration and efficiency.

S.no	Type of drive	Resultant speed(R_{res})
1.	Holonomic plus drive	2R
2.	Holonomic cross drive	$R_{res} = 2\sqrt{2}R$
3.	Steering drive	4R


Fig 1.2 chassis

List of Actuators and sensors used:-

- **Actuators used:**

S.No.	Component	Specifications	Required Power ($P = 2\pi NT/60$)	Usage
1	Orange Planetary geared DC motor	487 RPM, 12V DC, 43 N-cm (stall torque), 45.32W (rated power)	38W	Drives each wheel of the synergic four wheel steerable drive
2	E bike MY 1016Z3 motor (For Kicking)	324 RPM, 24V DC, 110 kg-cm (stall torque), 350W (rated power)	38.7W	Actuates the kicking mechanism
3	E bike MY 1016Z3 motor (For Passing)	324 RPM, 24V DC, 110 kg-cm (stall torque), 350W (rated power)	30.26W	Actuates the passing mechanism
4	Super Hercules motor	100 RPM, 12V DC, 150 kg-cm (stall torque)	More than sufficient	Rotates the wheels to orient the PR in any direction
5	Piston	50mm stroke length, ø16mm bore diameter	---	Grabs hold of the ball and releases it during passing
6	2 Pistons	100mm stroke length, Ø22mm bore diamemter	---	Part of a support mechanism used to stabilize the PR during Kicking

- **Sensors used:**

S.No.	Component	Specifications	Usage
1	Locomotion Encoder 1	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on the frontal side of the chassis, on a dummy wheel using suspension (to keep it in contact with the ground) to monitor the distance traversed by the robot
2	Locomotion Encoder 2	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on any of the wheel sprocket (except the dummy wheel) to monitor the direction of motion
3	Passing Mechanism Encoder	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on the passing mechanism to precisely determine the passing angle
4	Kicking Mechanism Encoder	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on the kicking mechanism to precisely determine the kicking angle
5	TF mini LIDAR	Max. Range 12m, 4.5-6V	2 LIDARS are attached on both sides of the robot to position it for the kick with respect to the wall
6	Class 2 Laser	Red point laser, 650nm wavelength	Used to align the PR with the TR to perform the passing of the Ball

Ball picking and passing mechanism:-

Picking the ball:-

A robotic arm with one degree of freedom is used such that the end effectors' opening is greater than the largest diameter of the ball in order to comfortably grab it. The ball grabbing mechanism uses a piston and the claw is covered with high friction material in order to hold and lift the ball without any slippage.

Passing the ball:-

The arm is attached to a sprocket which is driven by an E-bike motor .This motor provides power more than required to pass the ball (which is 30.26W according to our calculations for a distance of 4m). The motor is fixed on the lower part of the chassis and the power is transferred to the sprocket via a chain mechanism (which increases the stability by lowering the center of mass of the PR). The Pass Robot has a type B laser which is fixed at the frontal part of the robot. This laser is used by the Pass Robot for alignment with the Try Robot before the pass.

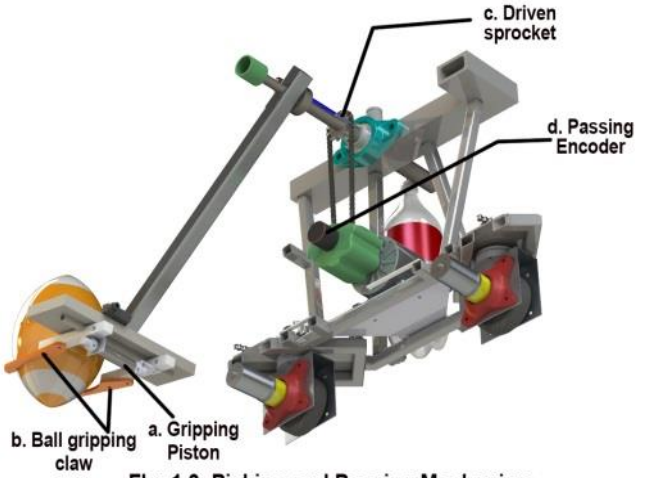


Fig 1.3 Picking and Passing Mechanism

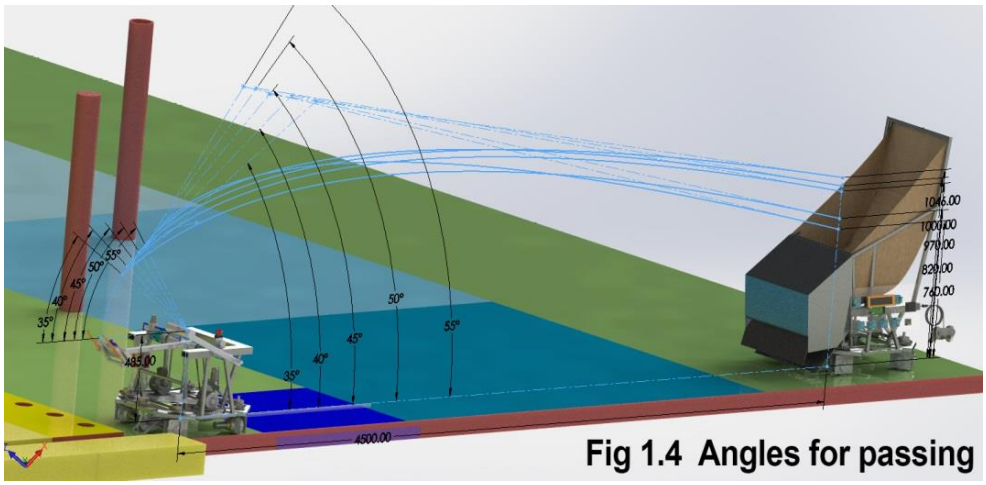


Fig 1.4 Angles for passing

When the robots are aligned, the passing mechanism starts to rotate and the ball is released with a velocity of 6.4 m/s, which when launched at an angle ranging from 32° to 55° ensures that the ball has a parabolic path and safely lands into the receiving mechanism of the Try Robot. An encoder is used to make sure that the ball is released within the range of these angles.

Kicking the ball:-

The kicking mechanism consists of a cylindrical "foot" (Fig 1.4.a), connected to a "leg" which is attached to a sprocket (Fig 1.4.b). This is also driven by a 24 V E-bike motor which also rests on the base of the chassis. The cylindrical block has been chosen for its convex contact area with the ball. It is solid so as to increase the mass of the foot, which in turn increases the moment and decreases any loss of energy in the form of plastic or elastic deformation. The kicking rod's length has been set as 416mm. The motor will be driven at approx. 260 rpm and thus the angular velocity of 27.16 rad/sec is achieved. From the above values, we obtain the tangential velocity of 11.36m/s.

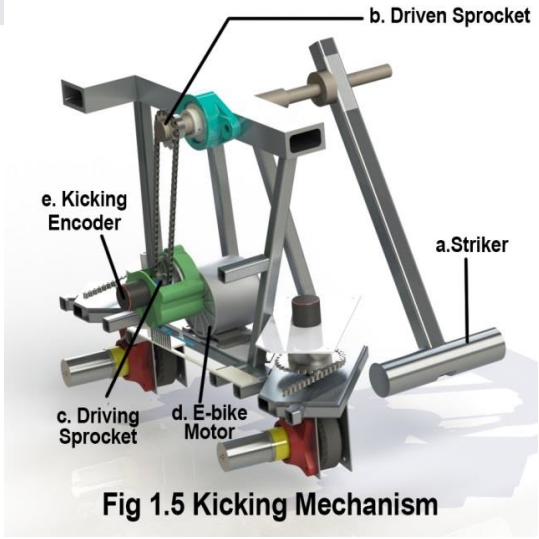


Fig 1.5 Kicking Mechanism

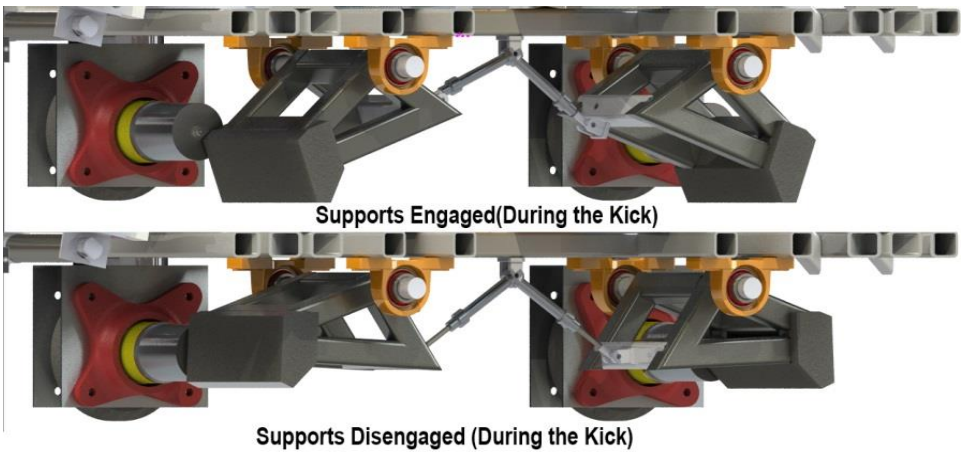


Fig 1.6 Braking Mechanism

To achieve a successful goal kick, the foot must hit the ball with an angle ranging from 37° to 62° for the above calculated tangential velocity. Also a power of 80W is required for the above kick, which is sufficiently supplied by the E-Bike motor (350W rated). While kicking, the orientation of the robot with respect to the wall is maintained using 2 TF mini LIDARs (applying PID algorithm) before every kick for higher precision and accuracy.

In order to maintain accuracy during the kick, the robot has to be stabilized using a support mechanism which is a first class lever actuated by a piston. When the member actuated by the piston touches the ground, it forms a simple wedge truss. This truss stabilizes the PR as any reaction force experienced by the PR is directed into the plate along the inclination of the wedge. Thus the PR doesn't experience any sudden jerks while kicking the ball.

Using the 2D projectile motion equations:

- In order to kick the ball through conversion post from KZ3,F the robot has to rotate at an angle of 22.3° W.R.T receiving zone wall
- Distance between the Tee (in KZ3) and the conversion post is 9723mm
- The ball has to pass through the area between the poles and has to be above a height of 1500mm for a goal
- When the ball is hit, its trajectory can be explained by taking the 2D projectile approach. The motion in the y-direction is continuously acted upon by the gravitational force and there is no force acting in the x-direction. Considering the following equations

$$where\ x = 9723mm,\quad u = 11.36m/s$$

$$x = u\cos\theta\,.t \qquad \text{------>1.}$$

$$t = \frac{x}{u\cos\theta} \qquad \text{------>2.}$$

$$y = u\sin\theta\,.t - \frac{1}{2}gt^2 \qquad \text{------>3.}$$

$$y = u\sin\theta\,.\frac{x}{u\cos\theta} - \frac{1}{2}g\frac{x}{u\cos\theta}^2 \qquad \text{------>4.}$$

$$y = x.\tan\theta - \frac{1}{2}g\frac{x}{u\cos\theta}^2 \qquad \text{------>5.}$$

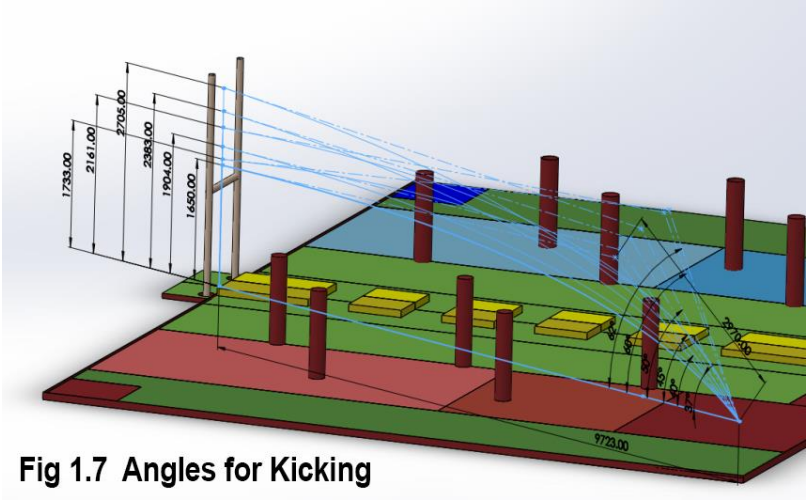


Fig 1.7 Angles for Kicking

- The final equation (Eqn 5.) is used to check if the height of the ball's trajectory is above 1500mm at the conversion post for different projection angles. Using the above equations we found that the angles ranging from 37° to 62° will successfully result in a goal
- Even if we take air resistance into account, the ball will be made to pass through the conversion post by slightly increasing the RPM to overcome it. The path of the trajectory is free of obstacles.

Design of Try Robot (TR):-

Specifications:

Length	920mm
Width	610mm
Height	1350mm
Weight	Approximately 18kg

Type of drive:-

The same synergic steering drive mechanism is implemented with the octagonal base as done in the Pass Robot. Four Tauren motors are used for the wheels' actuation and a super Hercules motor is used for steering the robot. An Arduino Mega is the brain of the system that receives commands from a PS2 controller that specifies the position and orientation.

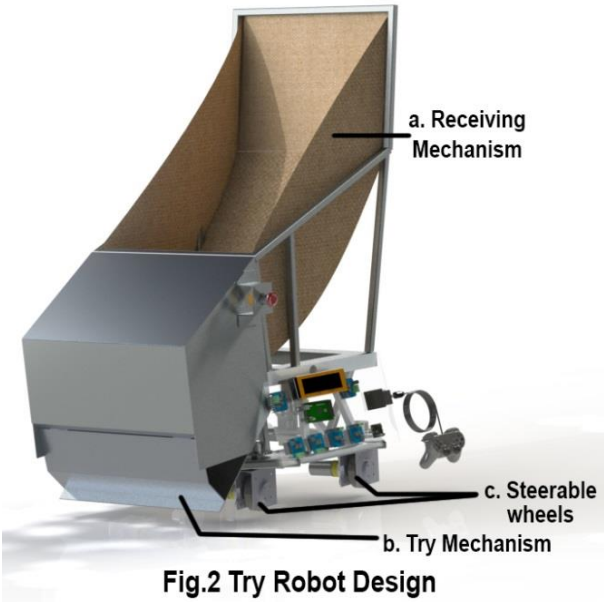


Fig.2 Try Robot Design

List of Actuators and sensors used:-

- **Actuators used:**

S.No.	Component	Specifications	Required Power ($P = 2\pi NT/60$)	Usage
1	Orange Planetary geared DC motor	487 RPM, 12V DC, 4.38 kg-cm (stall torque), 45.32W (rated power)	38W	Drives each wheel of the synergic four wheel steerable drive
2	Super Hercules motor	100 RPM, 12V DC, 150 kg-cm (stall torque)	More than sufficient	Rotates the wheels to orient the TR in any direction
3	Linear Actuator	12V DC,6mm/sec, 100mm stroke length	---	Actuates the Try mechanism in order to allow the ball to roll slowly into the try zone

- **Sensors used:**

S.No.	Component	Specifications	Usage
1	Locomotion Encoder 1	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on the frontal side of the chassis (on a dummy wheel) to monitor the distance traversed by the robot
2	Locomotion Encoder 2	Incremental type, 5-24V DC,PPR=360,CPR=1440	Attached on any wheel sprocket to monitor the direction of motion

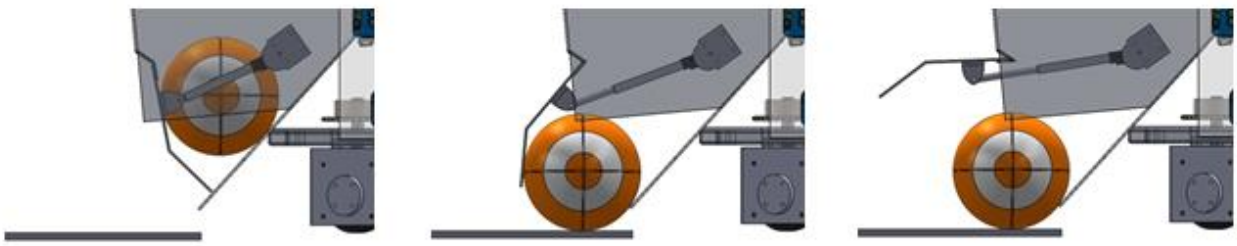
Ball receiving mechanism:-


Fig 2.1 Receiving Mechanism

The launched ball is received by the TR in a funnel like structure (Fig 2.1) which is covered by nylon mesh in order to damp the momentum of the ball which was accelerated to a speed of 6.4m/s. The funnel is extended in vertical axis i.e. 385mm in order to avoid a miss. The ball travels through the funnel and stops at the end of it due to the try plate and is now at a suitable position for the try task.

Try mechanism:-

The try mechanism (Fig 2.2) consists of a plate hinged to the upper casing and pivoted to a linear actuator (try plate) which is normally in contracted position. As soon as the TR reaches the try zone, the plate is pushed by the linear actuator i.e. the plate opens slowly and the ball rolls on the lower plate. The plate makes an angle of 47° with respect to the ground in order to decrease the rolling speed. The length of the plate over which the ball has to roll is 395mm so that the ball lies in contact with the try zone & the robot simultaneously as per the rules. This ensures that the ball is placed precisely in the try zone.


Fig 2.2 Try Mechanism